

户用储能电池包规格书

Product Specification

Battery Pack for Residential ESS

ERP356-L18-ST

版本 Version 1.0



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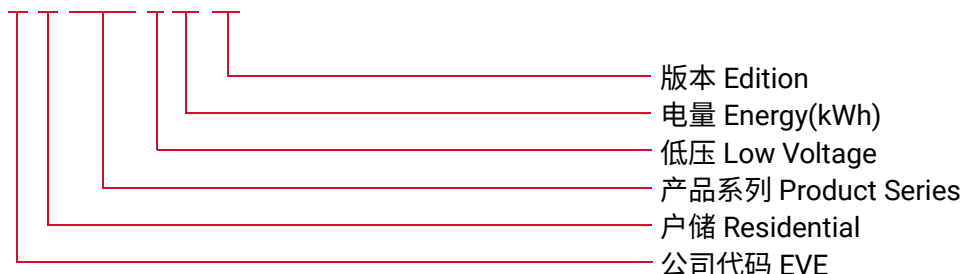
目 录 Contents

1 概述 Product Overview.....	1
2 产品特点 Product Features.....	2
3 产品规格 Product Specification	2
4 产品外观及结构 Product Appearance and Structure	4
5 包装运输要求 Packaging and Transportation Requirements.....	8
6 产品使用说明 Product Instructions.....	8
7 产品维护保养 Product Maintenance	9

1 概述 Product Overview

产品型号定义规则参考以下说明。The definition of product model is shown as follows.

E R P356-L18-ST



本规格书描述了该产品的结构设计、基本特性、测试方法和注意事项等，主要应用于户用储能应用场景。本产品的标准供货界面包含电池包及部分配件，不包含接地线以及电池包对外输出的高低压线束。详细供货清单参考下表。

The purpose of this document is to outline the structural design, key performance parameters, and safety precautions for the product, which is mainly applicated in residential ESS. The standard delivery scope of this product includes the battery pack and certain accessories, but excludes the grounding cable and the high-voltage and low-voltage wiring harnesses for external output from the battery pack. The details are as follows.

表 1 标准供货清单

Table 1 Standard Scope of Supply

序号 No.	组件 Component	型号 Model	数量 Quantity	备注 Remark
1	电池包 Battery pack	ERP356-L18-ST	1	
2	产品出厂检验报告 Test report	/	/	随产品出货 Shipped with product
3	产品合格证 Product certificate	/	/	随产品出货 Shipped with product
4	产品包装件 Product packaging	/	/	随产品出货 Shipped with product

注：具体型号规格和数量以供货前确认为准。

Note: The specific model, specification and quantity are subject to confirmation before delivery.

表 2 客户选配件清单(单个电池包)

Table 2 Customer optional accessories list (per battery pack)

序号 No.	组件 Component	描述 Description	数量 Quantity
1	正极输出动力线 Positive output power wires	OT 端子*2, 红色套管, 1000mm OT Terminal*2, red heat-shrink sleeving, 1000mm	1 set
2	负极输出动力线 Negative output power wires	OT 端子*2, 黑色套管, 1000mm OT Terminal*2, black heat-shrink sleeving, 1000mm	1 set
3	电池-逆变器通信线 Battery-inverter communication wire	双头 R J45 端子, 1000mm, 超五类屏蔽双绞线 Double-ended R J45 terminal, 1000mm, Cat.5e Shielded Twisted Pair	1 set
4	落地移动组件 Mobile floor assembly	福马轮+固定螺钉 Braking caster wheels + fastening screws	1 set
5	壁挂组件 Wall-mount assembly	塑胶站脚+壁挂支架+固定螺钉+膨胀螺栓 Plastic support feet + wall-mount bracket + fixing screws + expansion bolts	1 set
6	调试线 Debugging cable	通过电池包 RS232 接口 连接上位机, 用于售后调试 Connect to the host computer via the battery pack's RS232 interface for after-sales debugging	每个 20 尺集装箱 货柜至少配两根 At least two pieces per TEU

注：具体型号规格和数量以供货前确认为准；上述表格所列落地移动组件和壁挂组件为二者间任选其一。

Notes: The specific model, specification and quantity are subject to confirmation before delivery.

The mobile floor assembly and wall-mount assembly listed in the above table are mutually exclusive options.

2 产品特点 Product Features

- 小型化和易于安装
Compact size and easy installation
- 通过并联实现产品多样化, 最大支持 32 个电池包并联
Product diversification through parallel solutions(max.: 32 packs in parallel)
- 长循环和高安全性
Long cycle life and high safety
- 认证满足: UN38.3、IEC62619、IEC61000、欧盟新电池法、CEC 列名
Compliant with: UN38.3, IEC62619, IEC61000, Regulation(EU), CEC Listing

3 产品规格 Product Specification

表 3 电池包技术参数表

Table 3 Technical parameters (battery pack)

序号 No.	项目 Item		规格 Specification	备注 Remark
电芯 Cell	额定电压 Rated voltage		3.2V	
	额定容量 Rated capacity		353Ah	0.5P, 25°C ± 2°C, 2.5V~3.65V
	尺寸 T*W*H Dimension T*W*H		(72±0.5)*(186.6±0.5)*(211.6±0.5)mm	
	重量 Weight		6220g±150g	
电池包 Battery pack	成组方式 Configuration		1P16S	
	额定容量 Rated capacity		353Ah	
	额定电压 Rated voltage		51.2V	
	额定电量 Rated energy		18kWh	
	电压范围 Voltage range		43.2 V~58.4V	
	标准充/放电电流 Standard charge/discharge current		180A	0.5C, 25±2°C
	最大充/放电电流 Maximum charge/discharge current		200A	0.57C, 25±2°C
	推荐使用温度范围 Recommended temperature range for use		10°C~45°C	
	工作温度 Operating temperature	充电 Charge	0°C~60°C	可选配加热膜， 实现 0°C 以下充电 Optional heating film enables charging below 0°C.
		放电 Discharge	-20°C~60°C	
	工作湿度范围 Humidity range		5~85% RH, 无冷凝水 5~85% RH, no condensation	
	存储温度 Storage humidity	1 个月内 Within 1 month	-20°C~45°C	
		1 年内 Within 1 year	0°C~35°C	

	储存湿度 Storage humidity	<60% RH, 无冷凝水 <60% RH, no condensation
	尺寸 H*W*T Dimension H*W*T	(860±2)*(520±2)*(260±2)mm
	重量 Weight	133±2kg
	IP 等级 IP rating	IP65
	海拔高度 Application altitude	≤2000m

4 产品外观及结构 Product Appearance and Structure

4.1 产品外形以及尺寸 Product appearance and dimensions

电池包由 16 个 LF356L 电芯通过 1P16S 成组而成，其外形以及尺寸如下图。

The battery pack is consisted of 16 LF356L cells configured in 1P16S, and its appearance and dimensions are presented as follows.



图 1 电池包外观

Figure 1 The appearance of battery pack

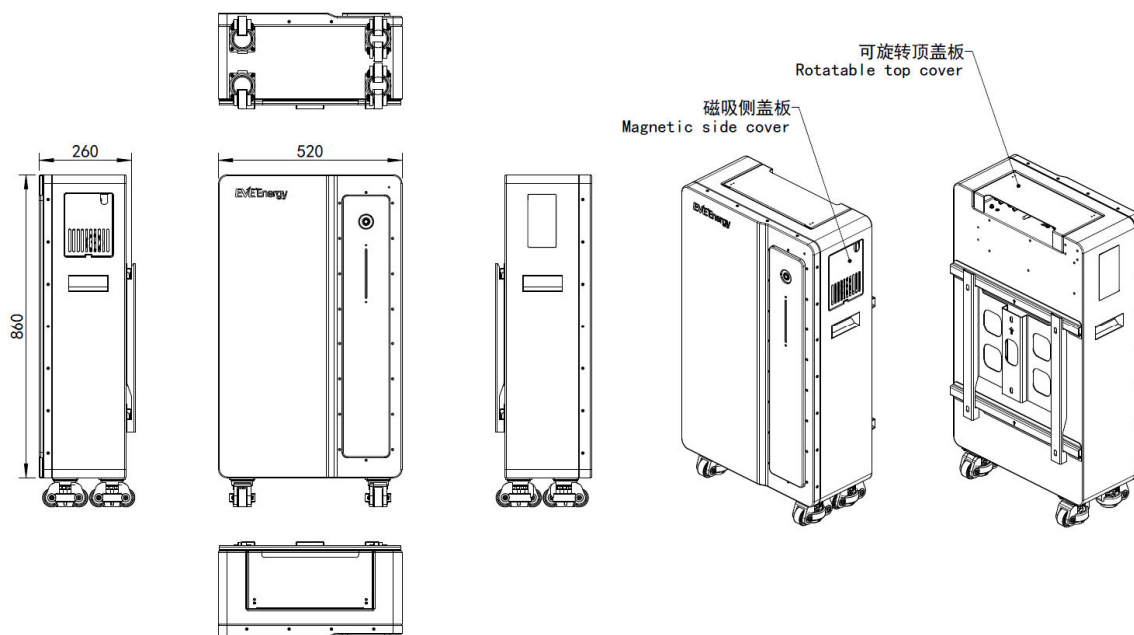


图 2 电池包尺寸图
Figure 2 The dimensions of battery pack

4.2 接口 Connection interface

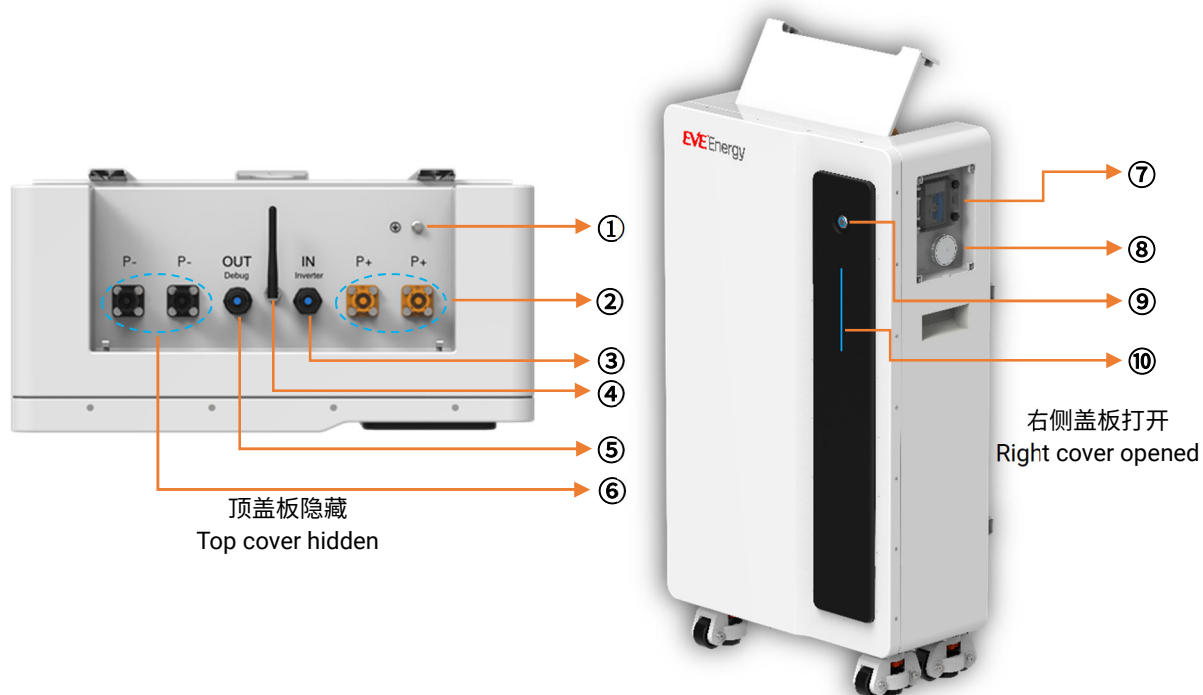


图 3 接口示意图
Figure 3 The diagram of connection terminals

表 4 对外接口表
Table 4 The table of external interface

序号 No.	项目 Item	描述 Description	
1	Grounding	接地点 Grounding point	
2	P+	电池正极 Battery positive	输出或者并机 Output or parallel connection
3	IN	并机通信输入、逆变器通信 (CAN/RS485) Parallel communication input, inverter communication (CAN/RS485)	详细定义见下表 See the table below for detailed definitions.
4	Antenna	天线 Antenna	
5	OUT	并机通信输出、调试口 Parallel communication output, debug port	详细定义见下表 See the table below for detailed definitions.
6	P-	电池负极 Battery negative	输出或者并机 Output or parallel connection
7	Breaker	断路器 Breaker	
8	Valve	泄压阀 Venting valve	
9	Switch	开关机按钮 Power ON/OFF button	
10	SOC	SOC 灯带 SOC light bar	

表 5 通信接口及定义表
Table 5 The definition of communication terminals

able to the definition of communication terminals

OUT/Debug			
引脚 Pin No.	定义 Definition	描述 Description	示意图 Diagram
1	RS485_B1	并机 485 通信 B 线 Parallel 485 communication (B line)	OUT Debug 
2	RS485_A1	并机 485 通信 A 线 Parallel 485 communication (A line)	
3	GND_ISO	通信隔离地 Communication isolated ground	
4	RS232_Tx	上位机 RS232 发送端 Host RS232 transmit	
5	RS232_Rx	上位机 RS232 接收端 Host RS232 receive	
6	DO	自动编址数字输出信号 Auto-addressing digital output	
7	RS485R+	多并机终端电阻接线，多机并联系统中， 首台与末台设备的该引脚需短接 Multi-parallel termination resistor connection. These pins must be shorted on the first and last units in a multi-parallel system.	
8	RS485R-		
IN/Inverter			
引脚 Pin No.	定义 Definition	描述 Description	示意图 Diagram
1	RS485_B1	并机 485 通信 B 线 Parallel 485 communication (B line)	IN Inverter 
2	RS485_A1	并机 485 通信 A 线 Parallel 485 communication (A line)	
3	GND_ISO	通信隔离地 Communication isolated ground	
4	CANH	PCS 通信 CANH PCS communication CANH	
5	CANL	PCS 通信 CAN 总线 L PCS communication CAN L	
6	DI	自动编址数字输入信号 Auto-addressing digital input	
7	RS485_A2	PCS 通信 485A 线 PCS communication 485 (A line)	
8	RS485_B2	PCS 通信 485B 线 PCS communication 485 (B line)	

注：具体接口型号及引脚定义以供货前确认为准。

The specific terminal model and pin definition are subject to confirmation before delivery.

5 包装运输要求 Packaging and Transportation Requirements

(1) 电池包包装方式：每个电池包采用单独的纸箱包装，纸箱内使用珍珠棉进行填充，保证电池包在运输过程中不晃动、安全可靠。

Packaging of a single Pack: Every single pack is packed in a single carton which is filled with pearl cotton to ensure that the packaged pack does not shake during transportation, making the product safe and reliable.

(2) 批量产品的包装采用缠绕膜+护角纸+木卡板/木箱打包方式。

The packaging of a large number of battery packs requires the use of wooden pallets or box with winding film, paper angle bead.

(3) 电池包应在 30%~50%SOC 的状态下进行运输。在运输过程中应防止剧烈振动、冲击或挤压、防止日晒雨淋。以上要求适用于陆运、海运运输方式。

The battery pack shall be transported at a state of charge (SOC) between 30% and 50%. During transportation, measures should be taken to prevent severe vibration, impact, or compression, as well as exposure to sunlight and rain. The above requirements apply to land and sea transportation modes.

6 产品使用说明 Product Instructions

(1) 请在使用前仔细阅读产品规格书，务必按照规定的充放电参数使用。

Please read the product instructions carefully before using the product, and be sure to use it according to the specified charge and discharge specifications.

(2) 禁止对电池包进行敲击、抛掷、踩踏。

Do not knock, throw or step on the battery pack.

(3) 禁止对电池包进行短路。

Do not short-circuit the battery pack.

(4) 电池包必须避免与腐蚀性物质接触，并远离火源及热源。禁止在阳光下存储或使用。

The battery pack must avoid contact with corrosive substances and be kept away from sources of fire and heat. Do not store or use it under the sun.

(5) 禁止将电池包投入水中。

Do not throw the battery pack into water.

(6) 禁止反向充电或放电。

Do not reversely charge or discharge the battery.

(7) 不支持新旧电池混用；

Mixing old and new batteries is not supported.

(8) 电池包必须在规定的条件下使用，过高或过低的环境温度都会影响电池的性能和安全。电池包必须在清洁、通风的环境条件下使用或存储。

The battery pack must be operated within the specified environment conditions, since too high or too low ambient temperature will affect the performance or safety of the battery. It must be used or stored in clean, ventilated environmental conditions.

(9) 电池包首次使用时，若有腐蚀，难闻气体或不正常现象，禁止使用。

Do not use the battery pack if there is corrosion, unpleasant smell or any abnormality during

the first usage.

(10) 电池包可以单独使用，也可以多个并联使用，电池包不能串联使用，电池包的并联数量不能超过 32。

The battery pack can be used alone or multiple in parallels. The battery packs cannot be used in series. The parallel quantity of the battery packs shall not exceed 32.

(11) 电池包必须在规定的充电倍率或功率条件下使用，并且充电上限电压不得超过产品技术要求，防止电池过充电，以免影响电池的充放电性能、机械性能和安全性能。

The battery pack must be operated within the specified charge conditions. Over-charge will lead the electrical performance, mechanical performance and safety performance to be decreased.

(12) 电池包禁止在强静电和强磁场的地方使用和存储，以免产生安全隐患。

Do not use or store the battery pack in places with strong static electricity and strong magnetic fields to avoid potential safety hazards.

(13) 未经我司允许，不得私自拆卸、重组或改装本产品，否则质保终止，若产生安全事故，我司概不负责。

This product shall not be disassembled or reassembled or modified without the permission of EVE, otherwise the warranty will be terminated, and our company will not responsible for any safety accidents.

(14) 电池包必须按照设计状态正立安装或存放，严禁侧放或倒置。

The battery pack must be installed or stored upright in accordance with the design state, and it is strictly forbidden to put it on its side or upside down.

电池包的使用必须严格遵照以上要求，否则质保终止，造成产品的性能损坏以及安全事故，我司概不负责。

The usage of battery pack must strictly comply with the above requirements, otherwise the warranty will be terminated, and EVE will not be responsible for any product performance damages or safety accidents.

7 产品维护保养 Product Maintenance

(1) 电池包长时间搁置或存放时，应保持在 30%~50%SOC 的状态下。

When the battery pack not in use for a long time, it is recommended to maintain the SOC at 30%~50%.

(2) 电池包长时间搁置或存放时，建议每 3 个月进行一次补充电，防止电池过放亏电，每半年应进行一次充放电循环。

When the battery pack is not in use for a long time, it shall be charged every 3 months regularly in case of the occurrence of over-discharge; And it is recommended to cyclically charge and discharge the battery pack every 6 months.

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